

Halvor Hafnor

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SUMMARY

Embedded software and ML engineer building sensor data classifiers for constrained hardware, from collection to on-device inference. Two years at Aerendir Mobile, a shipped automotive product, a pending patent, and an M.S. at UC San Diego.

EXPERIENCE

Embedded Software and ML Engineer

Aerendir Mobile · part time in term, full time out of term

Jul. 2024 – Present

Mountain View, CA

- Evaluated and hardened the motion-sensor classifiers behind **A-Live**, the anti-bot liveness check, and built the lab that measures how model and defense changes perform.
- Built Android applications for real-time biometric data streaming and workflow control.
- Collected and classified a 400 participant motion dataset for the company's biometric authentication work, each participant recorded from the accelerometer and gyroscope while sitting, standing and walking.
- Prototyped steering-wheel driver monitoring, shipped as **AerWheel**, with better than **99% accuracy** at under 0.5% false positives within 1.5 seconds, and named inventor on its pending patent. Trained scikit-learn decision trees on 240 Hz six-axis IMU features, run first on ST's on-sensor Machine Learning Core, then in firmware.

Electrical Engineering Intern

ABB Electrification

Jun. 2021 – Sep. 2023

Skien, Norway

- Assembled and wired around 300 industrial control panels and power cabinets from engineering schematics, across three consecutive summers.
- Supported production and testing of custom electrical systems for industrial clients.

EDUCATION

University of California - San Diego - San Diego, CA

M.S., Computer Engineering

Jan. 2026 – June 2027

B.S., Computer Engineering · Major GPA: 3.9

Sep. 2024 – June 2026

- **Provost Honors** every quarter enrolled, and **Tau Beta Pi**, the national engineering honor society.

De Anza College - Cupertino, CA

Computer Science & Engineering

GPA: 4.0

Sep. 2022 – June 2024

- Math & Science Tutor and CS Teaching Assistant for calculus, physics, discrete math, C++, Java and Assembly.

TECHNICAL SKILLS

Languages: C++, C, Python, TypeScript, Java, Bash, ARM Assembly

Embedded: Embedded Linux, Android, QML/Qt, STM32, STM32CubeIDE, ST sensor boards, firmware, on-sensor inference, Docker

Signals & ML: PyTorch, scikit-learn, NumPy, pandas, SciPy, OpenCV, YOLOv8, signal processing, feature extraction

Systems: Node, SQLite, WebSocket protocols

Agentic Development: Claude Code, MCP, spec-driven work breakdown, subagent orchestration, verification gates

PROJECTS

TabTerm, a macOS terminal in Chrome tabs

- A Chrome extension over a background terminal daemon that puts every shell session in a real Chrome tab, so tab groups, window management and keyboard shortcuts apply to terminals. Adds a command launcher, history search, project-scoped saved commands and a plugin API.

Laser targeting on an autonomous car, UC San Diego, ECE 148

- Real-time person tracking on a Raspberry Pi 5, in Docker. YOLOv8 detection and OAK-D stereo depth, converted through a 71.9° field of view into pan and tilt angles so two servos hold a laser on the nearest person from 0.5 to 2.9 m. Depth is the median of a 10x10 patch, since one bad pixel throws the aim.